

ASSOCIATION BETWEEN MICROVASCULAR ABNORMALITIES AND POST-EXERTIONAL MALAISE AMONG ADULTS DIAGNOSED WITH POSTURAL ORTHOSTATIC TACHYCARDIA SYNDROME (POTS)

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INTRODUCTION

Postural Orthostatic Tachycardia Syndrome (POTS) is a chronic autonomic nervous system disorder characterised by an excessive heart rate increase upon standing, without orthostatic hypotension. POTS predominantly affects women, with incidence rising since the COVID-19 pandemic, often precipitated by viral infection [1,2]. A "low-flow" POTS subtype has already been shown to exhibit impaired microvascular vasodilation using laser-Doppler flowmetry [3], confirming microvascular dysfunction in POTS, though direct skeletal muscle perfusion imaging has quantified this.

Post-exertional malaise (PEM) — a disproportionate worsening of symptoms following minimal exertion [4] — is significantly more common among myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) patients who also present with POTS or orthostatic hypotension than those who do not [5], reflecting substantial clinical overlap between the conditions, although PEM itself is not a diagnostic criterion of POTS. Near-infrared spectroscopy has shown that POTS patients display altered skeletal muscle oxygenation during head-up tilt compared with healthy controls [6], indicating orthostatic challenge may reveal deficits missed at rest. However, no study has quantified skeletal muscle microvascular perfusion using contrast-enhanced ultrasound (CEUS) — a validated technique for quantifying skeletal muscle microvascular blood flow [7] — under orthostatic stress in POTS, nor tested whether the degree of any abnormality relates to PEM severity.

This study aims to quantify the presence and degree of skeletal muscle microvascular perfusion abnormalities in adults with POTS, assessed via CEUS at rest and following active stand, and to determine whether these perfusion indices are associated with PEM severity.

METHODS

This cross-sectional pilot study will use a within-participant repeated-measures protocol to compare microvascular perfusion at rest and after orthostatic stress. Approximately 40 adults with confirmed POTS diagnosis will be recruited through partnership with Austin Health's Clinical Pharmacology Unit (Melbourne). This sample is powered to detect a moderate perfusion–PEM association ($r \approx 0.40$ – 0.45) at 80% power ($\alpha = 0.05$), balanced against feasibility given the specialised, multi-stage protocol. Exclusion criteria include peripheral arterial disease, diabetes with

microvascular complications, pregnancy, recent lower-limb injury, and ultrasound contrast agent hypersensitivity. Vasoactive medication use and time since diagnosis will be recorded as covariates rather than exclusion criteria, given the ethical and practical constraints of medication washout in a symptomatic population. Supine baseline CEUS of the gastrocnemius will be performed under standardised conditions (mid-morning, ≥ 3 hours fasted, having avoided caffeine, alcohol and strenuous exercise for 24 hours) [1], using a cuff-occlusion protocol [7], followed by repeat CEUS shortly after an active stand test, yielding three perfusion indices: microvascular blood volume, flow velocity, and blood flow. A subset ($n = 15$ – 20) will undergo distal leg skin biopsy for intraepidermal nerve fibre density (IENFD) to stratify neuropathic versus non-neuropathic POTS phenotypes [8]. PEM severity will be assessed via the DSQ-PEM-2, a self-report measure validated in ME/CFS populations but not separately in POTS [4,9]. Hierarchical regression will test whether post-stand perfusion predicts PEM severity beyond resting perfusion, while secondary analysis of the subsample will explore whether neuropathy status affects this relationship.

EXPECTED OUTCOMES AND SIGNIFICANCE

Microvascular perfusion deficits are expected to be present and associated with greater PEM severity. As exercise remains the primary recommended non-pharmacological POTS treatment [1], identifying a physiological marker of PEM risk could inform PEM-aware exercise prescription and clarify whether this risk relates to vascular versus neuropathic mechanisms.

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